

AWS Certified AI Practitioner AIF-C01

Section 9: AWS Managed AI Services

AWS Managed AI Services

AWS Managed AI Services are a suite of pre-built, fully managed machine learning tools that allow developers to inject intelligence into applications without writing machine learning code or training custom models.

AWS handles the underlying infrastructure, neural network architectures, and massive training datasets. You interact with these tools using simple API calls, software development kits (SDKs), or the AWS management console.

THIS APPROACH SUBSTANTIALLY REDUCES THE TIME TO VALUE. Developing an image-recognition, speech-recognition, translation, or natural-language-processing model internally may require acquiring training data, cleaning and labeling it, selecting algorithms, provisioning compute resources, training models, evaluating accuracy, deploying endpoints, monitoring performance, and retraining the models. A managed service packages much of this capability into an immediately consumable service.

MANAGED SERVICES ALSO PROVIDE ELASTIC SCALABILITY. A business might process a few hundred requests during normal operation and millions during a product launch. Services such as Rekognition and Lex can scale without the customer manually provisioning a fleet of inference servers.

ANOTHER BENEFIT IS REDUCED DEMAND FOR SPECIALIZED ML EXPERTISE. A software developer can call Amazon Translate or Polly without being an expert in neural machine translation or speech synthesis. This democratizes AI adoption, although it does not remove the need for domain knowledge, testing, security controls, responsible-AI practices, or human review.

AWS MANAGED SERVICES ALSO INTEGRATE WITH THE BROADER AWS ECOSYSTEM. Identity and authorization can be managed through IAM; data can be stored in S3; event-driven processing can use Lambda; encryption can use AWS Key Management Service; operational metrics can be observed in Amazon CloudWatch; and audit activity can be recorded through AWS CloudTrail. This integration makes the services easier to incorporate into secure, production-grade cloud architectures.

A FINAL ADVANTAGE IS THE SHIFT FROM LARGE UPFRONT INVESTMENT TO VARIABLE OPERATING EXPENSE. Instead of purchasing specialized hardware and maintaining an always-running AI environment, customers generally pay according to the relevant unit of consumption. However, the unit is not the same for every service. Some services charge by characters, requests, audio duration, images, video minutes, tokens, or compute time.

MANAGED DOES NOT MEAN THAT AWS ASSUMES EVERY RESPONSIBILITY. Under the AWS shared-responsibility model, the customer still needs to protect credentials, configure permissions, classify and secure data, select appropriate Regions, comply with legal requirements, test model performance, examine bias and limitations, monitor outputs, and implement human oversight for consequential decisions.

1. Amazon Comprehend

AMAZON COMPREHEND IS A MANAGED NATURAL LANGUAGE PROCESSING (NLP) SERVICE USED TO UNDERSTAND AND ANALYZE TEXT.

It can identify entities such as people, organizations and locations, extract key phrases, detect the dominant language, detect personally identifiable information (PII), and determine sentiment as positive, negative, neutral, or mixed. **AMAZON COMPREHEND PERFORMS NLP TASKS SUCH AS SENTIMENT ANALYSIS, ENTITY DETECTION, AND KEY-PHRASE EXTRACT.**

tion. For the exam, associate Comprehend with “**understand text.**” For example, if a company wants to analyze thousands of customer reviews to determine whether customers are happy or unhappy, Amazon Comprehend is a strong fit.

Exam memory:

Comprehend = understand/analyze text.

2. Amazon Translate

AMAZON TRANSLATE IS A MANAGED MACHINE-TRANSLATION SERVICE THAT CONVERTS TEXT FROM ONE SUPPORTED LANGUAGE TO ANOTHER. It is useful for multilingual websites, customer-support systems, document translation and application localization.

Translate works with **text**, not directly with speech. If you receive spoken Spanish and want English text, for example, you could first use Amazon Transcribe to convert speech to text and then Amazon Translate to translate that text.

Exam memory:

Translate = language A → language B.

3. Amazon Transcribe

AMAZON TRANSCRIBE PERFORMS AUTOMATIC SPEECH RECOGNITION (ASR). IT CONVERTS SPOKEN AUDIO INTO WRITTEN TEXT and supports both real-time streaming transcription and batch transcription of recorded files. It can also distinguish speakers, work with multiple audio channels, use custom vocabularies and filter sensitive or unwanted content.

Typical applications include meeting transcripts, subtitles, contact-center recordings, voice applications and searchable audio archives.

The key exam distinction is:

Transcribe = speech → text.

Do not confuse it with Amazon Polly, which performs the opposite transformation.

4. Amazon Polly

AMAZON POLLY CONVERTS WRITTEN TEXT INTO NATURAL-SOUNDING SPEECH. IT IS THEREFORE A TEXT-TO-SPEECH SERVICE. Applications include audiobooks, accessibility tools, spoken notifications, virtual assistants and applications that read content aloud.

Polly supports multiple voices and languages, and AWS provides different speech-generation technologies depending on the selected voice.

For the exam:

Polly = text → speech.

A simple way to remember the pair is:

Transcribe listens. Polly speaks.

5. Amazon Rekognition

AMAZON REKOGNITION IS A MANAGED COMPUTER-VISION SERVICE USED TO ANALYZE IMAGES AND VIDEOS.

It can detect objects, scenes, concepts, text, faces and potentially unsafe content, and can perform face comparison for supported use cases.

For example, a photo-sharing application could automatically identify cars, animals and buildings in uploaded images. A content platform could use Rekognition to flag potentially inappropriate images for further review.

For the exam, associate Rekognition with:

“What is inside this image or video?”

DO NOT CONFUSE REKOGNITION WITH Textract. REKOGNITION ANALYZES VISUAL CONTENT, WHEREAS Textract SPECIALIZES IN DOCUMENTS.

6. Amazon Lex

AMAZON LEX V2 IS USED TO BUILD CONVERSATIONAL INTERFACES AND CHATBOTS USING VOICE AND TEXT.

It combines automatic speech recognition with natural-language understanding so that applications can understand what a user wants and conduct a structured conversation.

Important Lex concepts include **intents, utterances and slots**.

- An intent represents what the customer wants to do, such as BookHotel.
- An utterance is something the user might say, such as “I want to reserve a room.”
- Slots are pieces of information required to complete the request, such as city, date and number of guests.

LEX COMMONLY INTEGRATES WITH AWS LAMBDA to execute business logic after collecting required information.

Exam memory:

Lex = conversational chatbot/voice bot.

7. Amazon Textract

AMAZON Textract EXTRACTS INFORMATION FROM SCANNED DOCUMENTS AND IMAGES. Unlike basic OCR, Textract can identify **typed text, handwriting, forms, key-value pairs, tables, invoices, receipts and supported identity documents.**

For example, an insurance company could upload thousands of claim forms and use Textract to extract customer names, policy numbers, dates and table values automatically.

The main exam distinction is:

Rekognition → understand images.

Textract → understand documents.

If the question mentions **forms, invoices, receipts, tables or scanned business documents**, think Textract.

8. Amazon Personalize

AMAZON PERSONALIZE IS A FULLY MANAGED ML SERVICE USED TO CREATE PERSONALIZED RECOMMENDATIONS BASED ON USER BEHAVIOR AND BUSINESS DATA. It can generate item recommendations and user segments based on users' affinities for particular products or content.

Typical use cases include:

“Recommended for you”, “Customers who viewed this also viewed...”, personalized movies, personalized news feeds and e-commerce product recommendations.

For the exam, associate:

Personalize = recommendation engine.

A streaming company wanting to recommend movies based on viewing history is a classic Amazon Personalize scenario.

9. Amazon Kendra

AMAZON KENDRA IS A MANAGED INTELLIGENT ENTERPRISE SEARCH AND INFORMATION-RETRIEVAL SERVICE. Instead of relying only on keyword matches, Kendra uses semantic and contextual understanding to return documents and passages relevant to natural-language questions.

For example, employees could search company manuals, internal documents or knowledge repositories by asking questions in natural language. Kendra can connect multiple enterprise data repositories and create a unified searchable index.

Kendra can also provide retrieval capabilities for generative AI and RAG architectures, including integrations with Amazon Bedrock knowledge bases.

Exam memory:

Kendra = intelligent enterprise search.

10. Amazon Mechanical Turk

AMAZON MECHANICAL TURK (MTurk) IS A CROWDSOURCING MARKETPLACE THAT CONNECTS ORGANIZATIONS WITH HUMAN WORKERS WHO PERFORM SMALL TASKS KNOWN AS MICROTASKS. Examples

include image annotation, content categorization, data validation, surveys and tasks where human judgment is preferable to automated processing.

The important conceptual difference is that Mechanical Turk provides a **human workforce**, not an AI model.

Exam memory:

Mechanical Turk = humans performing tasks at scale.

11. Amazon Augmented AI — A2I

AMAZON AUGMENTED AI (AMAZON A2I) ADDS HUMAN REVIEW TO MACHINE-LEARNING PREDICTIONS. A common pattern is to let the AI handle high-confidence predictions automatically but send low-confidence or selected predictions to human reviewers.

For example, Textract may confidently extract most values from an application form. If its confidence in the applicant's name is low, A2I can trigger a human review workflow. A2I has also supported human-review workflows involving Rekognition and custom ML predictions.

A current product-status distinction is important: **Amazon SageMaker A2I is no longer open to new customers. Existing customers can continue using it, but AWS states that it does not plan to introduce new features.**

For exam concepts, remember:

A2I = AI + human review loop.

12. Amazon Comprehend Medical

Amazon Comprehend Medical applies NLP to **medical and clinical text**. **IT CAN IDENTIFY MEDICAL CONDITIONS, MEDICATIONS, TREATMENTS, TESTS, PROCEDURES AND PROTECTED HEALTH INFORMATION (PHI) FROM UNSTRUCTURED MATERIAL SUCH AS PHYSICIAN NOTES, DISCHARGE SUMMARIES AND TEST RESULTS.**

It can also link detected medical concepts with standardized medical ontologies such as **ICD-10-CM, RxNorm and SNOMED CT**.

Think of the distinction as:

Comprehend → general text NLP.

Comprehend Medical → medical text NLP.

13. Amazon Transcribe Medical

AMAZON TRANSCRIBE MEDICAL IS SPECIALIZED SPEECH RECOGNITION FOR HEALTHCARE. It converts medically related speech into text and is designed for scenarios such as physician dictation, telemedicine sessions and clinician-patient conversations. It supports both streaming and batch transcription.

The core distinction is:

Transcribe → general speech → text.

Transcribe Medical → medical speech → text.

14. AWS HealthScribe

AWS HEALTHSCRIBE COMBINES SPEECH RECOGNITION AND GENERATIVE AI FOR HEALTHCARE CONVERSATIONS. It can transcribe clinician-patient discussions, identify speaker roles, identify medical terms and create preliminary clinical notes that clinicians can review.

This makes HealthScribe broader than simple medical transcription.

A useful comparison is:

Transcribe Medical → medical speech → transcript.

Comprehend Medical → medical text → structured medical information.

HealthScribe → clinician-patient conversation → transcript + medical information + draft clinical notes.

AWS HealthScribe is HIPAA eligible and is intended to help healthcare software providers build clinical documentation applications.

15. Amazon EC2 for AI and ML

AMAZON EC2 PROVIDES VIRTUAL SERVERS CALLED INSTANCES IN THE AWS CLOUD. Unlike specialized services such as Rekognition or Translate, EC2 gives organizations greater control over the actual computing environment. For AI workloads, customers can select instances with CPUs, GPUs or specialized AWS machine-learning accelerators.

The AIF-C01 distinction is important: use managed AI services when AWS already provides the capability you require; consider EC2 when you need **greater infrastructure control**, specialized hardware, custom frameworks or custom model-training/inference environments.

EC2 GPU Instances

GPUs perform many mathematical operations in parallel, which makes them particularly useful for deep learning and generative-AI workloads. AWS provides multiple accelerated EC2 instance families containing NVIDIA and other accelerators. Current EC2 families include GPU-based options such as G and P families, although exact generations evolve over time.

In broad exam terms:

CPU → normal/general computing.

GPU → highly parallel AI/ML computation.

GPUs are frequently used for both **training and inference**, especially for large deep-learning models.

AWS Trainium

AWS TRAINIUM IS AN AWS-DESIGNED ACCELERATOR OPTIMIZED PRIMARILY FOR HIGH-PERFORMANCE DEEP-LEARNING TRAINING. EC2 Trn instance families use Trainium accelerators and can work with popular ML frameworks through the AWS Neuron software stack.

For exam questions:

Trainium → TRAIN models.

That name association is useful and generally captures the intended distinction.

AWS Inferentia

AWS INFERENTIA IS AN AWS-DESIGNED ACCELERATOR OPTIMIZED FOR MACHINE-LEARNING INFERENCE. Inference means running a trained model to generate predictions or outputs. Inferentia-powered EC2 instances are designed for high-performance, low-latency model inference.

For the exam:

Inferentia → INFERENCE.

Therefore:

Trainium → training

Inferentia → inference

EC2 Auto Scaling Groups — ASG

An **Auto Scaling Group (ASG)** automatically adjusts the number of EC2 instances according to workload demand and configured scaling policies.

Imagine an ML inference application normally requires two servers. During a sudden spike in requests, Auto Scaling can launch additional instances. When traffic falls, it can terminate unnecessary instances.

For the exam:

ASG = automatically scale EC2 capacity up or down.

This provides better scalability, availability and cost efficiency.

Elastic Load Balancing — ELB

Elastic Load Balancing (ELB) distributes incoming traffic across multiple healthy targets such as EC2 instances. This prevents one instance from receiving all requests while others remain underutilized.

ASG and ELB commonly operate together:

ELB → distributes traffic.

ASG → adjusts the number of instances.

AWS SPECIFICALLY SUPPORTS CONNECTING AUTO SCALING GROUPS WITH ELASTIC LOAD BALANCING SO TRAFFIC CAN BE DISTRIBUTED ACROSS HEALTHY INSTANCES, INCLUDING DEPLOYMENTS ACROSS MULTIPLE AVAILABILITY ZONES.

An AI inference architecture could therefore look conceptually like:

Users → Load Balancer → EC2 inference instances → ML model

while:

Auto Scaling Group → adds/removes EC2 instances according to demand.

Reference Architecture for a Scalable AI Inference Endpoint

Layer	AWS Component	Role
Traffic entry	Application Load Balancer	Accepts requests and routes them to healthy model servers.
Compute fleet	EC2 GPU, Trainium, Inferentia, or CPU instances	Hosts the model and performs inference.
Elastic capacity	Auto Scaling group	Adds, removes, and replaces instances according to demand and health.
Model and data storage	Amazon S3 and/or Amazon EBS	Stores model artifacts, configuration, datasets, and persistent volumes.
Monitoring	Amazon CloudWatch	Tracks latency, throughput, errors, resource utilization, and scaling signals.
Access control	IAM, security groups, and VPC controls	Restricts service permissions and network access.

Scenario-Based AIF-C01 Questions

Scenario 1 — Customer Sentiment

A company receives 300,000 written product reviews every month. It wants to determine whether each review expresses positive, negative, neutral or mixed sentiment without developing its own NLP model.

Which service should it use?

- A. Amazon Personalize
- B. Amazon Comprehend
- C. Amazon Kendra
- D. Amazon Translate

Answer: B — Amazon Comprehend

Why: Sentiment analysis of text is a core Comprehend capability.

Scenario 2 — Audio Translation

A multinational company receives recorded Spanish customer-support calls. It needs an English text version of each conversation.

Which combination is appropriate?

- A. Polly → Translate
- B. Transcribe → Translate
- C. Translate → Transcribe
- D. Rekognition → Translate

Answer: B — Transcribe → Translate

First:

Audio → Transcribe → Spanish text

Then:

Spanish text → Translate → English text

Scenario 3 — Accessibility

A news company wants visually impaired users to hear articles from its website.
Which service is most appropriate?

- A. Amazon Transcribe
- B. Amazon Polly
- C. Amazon Comprehend
- D. Amazon Lex

Answer: B — Amazon Polly

The articles already exist as text, so Polly converts the text into speech.

Scenario 4 — Invoice Processing

A financial organization receives thousands of scanned invoices. It needs to extract supplier names, invoice numbers and table values.

Which service should it choose?

- A. Amazon Rekognition
- B. Amazon Kendra
- C. Amazon Textract
- D. Amazon Personalize

Answer: C — Amazon Textract

The important clues are **scanned documents, structured fields and tables**.

Scenario 5 — Video Recommendations

A streaming service wants to recommend movies based on each user's historical viewing behaviour.

Which service is designed for this requirement?

- A. Amazon Personalize
- B. Amazon Rekognition
- C. Amazon Comprehend
- D. Amazon Kendra

Answer: A — Amazon Personalize

The clue is **personalized recommendation**.

Scenario 6 — Employee Knowledge Search

A company stores thousands of HR manuals, policies, technical documents and support guides.

Employees want to search them using questions such as:

“How many vacation days do new employees receive?”

Which service is the strongest fit?

- A. Amazon Kendra
- B. Amazon Textract
- C. Amazon Rekognition
- D. Amazon Translate

Answer: A — Amazon Kendra

The requirement is **semantic enterprise search across organizational information**.

Scenario 7 — Human Review

A bank uses machine learning to process financial forms. Predictions with confidence above 95% can proceed automatically, but lower-confidence predictions need a human to verify them.

Which concept/service best matches the requirement?

- A. Amazon Polly
- B. Amazon A2I
- C. Amazon Kendra
- D. Amazon Personalize

Answer: B — Amazon A2I

The key phrase is **human review of low-confidence ML predictions**.

For current deployments, remember that SageMaker A2I is no longer open to new customers; this scenario remains useful for understanding the exam concept.

Scenario 8 — Medical Notes

A healthcare application already has thousands of written clinical notes. It needs to identify medications, dosages, medical conditions and PHI.

Which service should it use?

- A. Amazon Transcribe Medical
- B. Amazon Comprehend Medical
- C. AWS HealthScribe
- D. Amazon Polly

Answer: B — Amazon Comprehend Medical

The input is already **clinical text**, so medical NLP—not transcription—is required.

Scenario 9 — Doctor-Patient Documentation

A healthcare software company wants to record doctor-patient conversations and automatically produce transcripts and preliminary clinical notes for clinician review.

Which service best fits the requirement?

- A. Amazon Transcribe
- B. Amazon Comprehend
- C. AWS HealthScribe
- D. Amazon Lex

Answer: C — AWS HealthScribe

The distinguishing requirement is not simply transcription; it includes **generating draft clinical documentation from the conversation**.

Scenario 10 — High-Traffic AI Inference

A company deploys a custom ML model on Amazon EC2. Traffic changes considerably throughout the day. The company needs additional inference servers to start automatically as requests increase and wants incoming requests distributed across the healthy instances.

Which combination should it use?

- A. Amazon Textract + Amazon Kendra
- B. Trainium + Amazon Translate
- C. Auto Scaling Group + Elastic Load Balancing
- D. Amazon Comprehend + Amazon Lex

Answer: C — Auto Scaling Group + Elastic Load Balancing

The responsibilities are different:

ASG → changes capacity.

ELB → distributes traffic.

Scenario 11 — Custom Model Training

A research team needs to train a very large deep-learning model on EC2 and wants to use an AWS-designed accelerator optimized for training instead of general-purpose CPUs.

Which accelerator should it consider?

- A. AWS Inferentia
- B. AWS Trainium
- C. Elastic Load Balancing
- D. Amazon Kendra

Answer: B — AWS Trainium

Remember:

Trainium → training.

Scenario 12 — Cost-Efficient ML Inference

A company has already trained its neural-network model. It now wants an AWS-designed EC2 accelerator for high-performance production predictions with low latency.

Which accelerator is designed for this stage?

- A. Trainium
- B. Inferentia
- C. Mechanical Turk
- D. Amazon Textract

Answer: B — AWS Inferentia

The model is **already trained**. The workload is therefore **inference**, making Inferentia the intended answer.